

1 Research Questions

The empirical analysis is organized around a semi-structural state-space representation of the HSA mechanism. The objective is not to estimate the complete structural model or to recover the underlying HSA demand system. Instead, the analysis asks whether the central empirical implications of the HSA Phillips curve are supported by U.S. aggregate data.

Throughout this section, the competition measure is oriented so that a larger value of N_t represents a more competitive environment, or equivalently a larger effective number of firms. Hence, a decline in N_t corresponds to greater market concentration. If a concentration measure such as an HHI is used directly instead, the directional signs stated below must be reversed.

Let the observed competition measure admit the state-space representation

$$N_t^{\text{obs}} = \bar{N}_t + \hat{N}_t + \nu_t^N, \quad (1)$$

where $\bar{N}_t$ denotes a slow-moving competitive environment, $\hat{N}_t$ denotes a stationary latent deviation around that environment, and ν_t^N denotes measurement error.

For interpretation and reporting, define

$$\tilde{N}_t = \bar{N}_t - N_0, \quad (2)$$

where N_0 is a fixed reporting center.

The empirical Phillips curve is represented as

$$\pi_t = \alpha E_t \pi_{t+1} + \kappa_t x_t + \theta_t \hat{N}_t + \varepsilon_t, \quad (3)$$

where x_t denotes the Phillips-curve forcing variable. The two competition-dependent coefficients are parameterized locally as

$$\kappa_t = \kappa_0 + \delta \tilde{N}_t, \quad (4)$$

and

$$\theta_t = \theta_0 + \gamma \tilde{N}_t. \quad (5)$$

The coefficients κ_t and θ_t are empirical varying coefficients. Their structural interpretation is conditional on the maintained mapping between the observed competition measure, the latent competition states, and the competitive environment in the HSA model.

The directional implications below are not imposed mechanically in the baseline estimation. The parameters are estimated without sign restrictions, and the posterior probability assigned to the theory-consistent region is reported as an empirical measure of mechanism consistency.

1.1 RQ1: Competition-Dependent Phillips-Curve Slope

Does the conditional response of inflation to the forcing variable vary systematically with the competitive environment?

The first research question concerns the slow-moving component of the Phillips-curve slope. Equation (4) allows the conditional slope to vary with the competitive environment:

$$\kappa_t = \kappa_0 + \delta \tilde{N}_t. \quad (6)$$

The parameter δ measures how the conditional Phillips-curve slope changes with the slow-moving competition state.

Under the maintained HSA interpretation, greater market concentration reduces the structural slope of the Phillips curve. Since larger values of N_t are defined here to represent greater competition, the directional hypothesis is

$$H_0^{(1)} : \delta \leq 0 \quad \text{against} \quad H_1^{(1)} : \delta > 0. \quad (7)$$

Thus, a positive δ implies that the Phillips curve becomes steeper when competition increases and flatter when market concentration rises. In the Bayesian analysis, the corresponding posterior quantity of interest is

$$\Pr(\delta > 0 \mid \mathcal{D}), \quad (8)$$

where $\mathcal{D}$ denotes the observed data.

1.2 RQ2: Short-Run Direct Competition Channel

Does the stationary competition state contain inflation-relevant information beyond the standard Phillips-curve forcing variable?

The second research question concerns the direct loading of inflation on the stationary competition state. Holding the interaction in Equation (5) fixed at the reporting center gives

$$\pi_t = \alpha E_t \pi_{t+1} + \kappa_t x_t + \theta_0 \hat{N}_t + \varepsilon_t. \quad (9)$$

The coefficient θ_0 measures the direct short-run association between the stationary competition state and inflation after conditioning on expected inflation and the Phillips-curve forcing variable.

In the HSA mechanism, endogenous movements in the number of firms generate a cost-push component distinct from the conventional marginal-cost term. With N_t oriented so that larger values correspond to greater competition, the theory-consistent direct loading is negative. The directional hypothesis is therefore

$$H_0^{(2)} : \theta_0 \geq 0 \quad \text{against} \quad H_1^{(2)} : \theta_0 < 0. \quad (10)$$

The corresponding posterior quantity is

$$\Pr(\theta_0 < 0 \mid \mathcal{D}). \quad (11)$$

In the baseline empirical interpretation, θ_0 is treated conservatively as a direct latent-state loading. Its interpretation as the endogenous-entry or participation channel of the HSA model is conditional on the maintained theoretical mapping.

1.3 RQ3: Competition-Dependent Dynamic Effect

Does the strength of the short-run competition channel itself vary with the slow-moving competitive environment?

The third research question asks whether the direct competition loading is itself state dependent:

$$\theta_t = \theta_0 + \gamma \tilde{N}_t. \quad (12)$$

The coefficient γ measures how the short-run effect associated with $\hat{N}_t$ changes as the economy moves across different competitive environments.

Under the maintained HSA interpretation with the relevant pass-through channel, greater concentration strengthens the magnitude of the negative endogenous competition effect. Since lower values of N_t represent greater concentration, this implies that θ_t should become more negative as $\tilde{N}_t$ falls. In the linear approximation above, the resulting directional hypothesis is

$$H_0^{(3)} : \gamma \leq 0 \quad \text{against} \quad H_1^{(3)} : \gamma > 0. \quad (13)$$

A positive γ therefore implies

$$\tilde{N}_t \downarrow \implies \theta_t \downarrow, \quad (14)$$

so that the direct short-run competition channel becomes more negative as market concentration increases.

The corresponding posterior quantity is

$$\Pr(\gamma > 0 \mid \mathcal{D}). \quad (15)$$

This prediction is stronger than the existence of the direct effect itself. The sign of θ_0 is motivated by the endogenous cost-push channel, whereas the directional prediction for γ additionally requires that the magnitude of that channel varies systematically with the competitive environment.

1.4 RQ4: Joint HSA Mechanism Consistency

Are the slope and dynamic competition channels jointly organized in the direction implied by the HSA mechanism?

The fourth research question evaluates the mechanism jointly rather than testing each coefficient in isolation. The empirical Phillips curve contains two conceptually distinct competition channels:

$$\underbrace{\kappa_t x_t}_{\text{competition-dependent slope channel}} \quad \text{and} \quad \underbrace{\theta_t \hat{N}_t}_{\text{direct dynamic competition channel}}. \quad (16)$$

The first channel captures systematic variation in the conditional Phillips-curve slope across competitive environments. The second captures a direct inflation contribution from short-run competition dynamics that is not absorbed by the conventional forcing variable.

For the orientation of N_t maintained above, the full theory-consistent sign configuration is

$$\delta > 0, \quad \theta_0 < 0, \quad \gamma > 0. \quad (17)$$

Rather than reducing this multidimensional restriction to a single coefficient test, the Bayesian analysis evaluates the joint posterior probability

$$\Pr(\delta > 0, \theta_0 < 0, \gamma > 0 \mid \mathcal{D}). \quad (18)$$

This quantity measures how much posterior probability is assigned to the region in which all three directional implications of the empirical HSA mechanism hold simultaneously.

The analysis also reports the three marginal directional probabilities

$$\Pr(\delta > 0 \mid \mathcal{D}), \quad \Pr(\theta_0 < 0 \mid \mathcal{D}), \quad \Pr(\gamma > 0 \mid \mathcal{D}), \quad (19)$$

so that a failure of joint mechanism consistency can be traced to a particular empirical channel.

1.5 Nested Model Interpretation

The research questions generate a nested sequence of empirical specifications:

$$\text{Model 1:} \quad \delta = \theta_0 = \gamma = 0, \quad (20)$$

$$\text{Model 2:} \quad \delta \text{ free,} \quad \theta_0 = \gamma = 0, \quad (21)$$

$$\text{Model 3:} \quad \delta, \theta_0 \text{ free,} \quad \gamma = 0, \quad (22)$$

$$\text{Model 4:} \quad \delta, \theta_0, \gamma \text{ free.} \quad (23)$$

Model 1 is the competition-invariant benchmark. It allows a conventional Phillips-curve relationship but rules out both competition channels.

Model 2 introduces the competition-dependent slope channel and therefore addresses RQ1.

Model 3 additionally introduces the direct stationary competition channel and therefore addresses RQ2 while retaining the slope channel from Model 2.

Model 4 allows the direct competition coefficient itself to vary with the slow-moving competitive environment and therefore addresses RQ3 and the full joint mechanism in RQ4.

The sequence can therefore be summarized as

$$\text{Competition-Invariant} \longrightarrow \text{Slope Channel} \longrightarrow \text{Slope} + \text{Direct Channel} \longrightarrow \text{Full State-Dependent HSA}. \quad (24)$$

Model comparison is based on posterior parameter distributions, marginal likelihoods, and predictive performance. The purpose is not only to determine whether a time-varying Phillips curve fits the data better, but also to determine whether its time variation is systematically linked to the competitive environment in the direction implied by the HSA mechanism.

1.6 Mechanism Decomposition

Conditional on the estimated state paths, Equation (3) permits the competition-related contribution to inflation to be separated into two components.

The slope channel is

$$C_t^{\text{slope}} = (\kappa_t - \kappa_0) x_t = \delta \tilde{N}_t x_t, \quad (25)$$

while the direct dynamic channel is

$$C_t^{\text{dynamic}} = \theta_t \hat{N}_t = \left(\theta_0 + \gamma \tilde{N}_t \right) \hat{N}_t. \quad (26)$$

Their sum,

$$C_t^{\text{HSA}} = C_t^{\text{slope}} + C_t^{\text{dynamic}}, \quad (27)$$

provides an empirical decomposition of the competition-related component of fitted inflation.

This decomposition distinguishes a change in the conditional transmission of the forcing variable from a separate short-run competition contribution. It therefore provides the empirical basis for assessing whether the observed flattening is primarily associated with a changing Phillips-curve slope, a distinct dynamic competition channel, or both.

2 Case 1: Quarterly Joint Estimation with Capital IQ Competition Data

Capital IQ provides a quarterly competition measure, so inflation and competition are jointly observed at the quarterly frequency. The competition series is decomposed into a slow-moving component and a stationary cyclical component.

$$\mathbf{s}_t = \begin{bmatrix} \tilde{N}_t \\ \hat{N}_t \end{bmatrix}, \quad \tilde{N}_t = \bar{N}_t - N_0, \quad n_t^{\text{CIQ}} = N_t^{\text{CIQ}} - N_0. \quad (28)$$

The transition equation is

$$\mathbf{s}_t = \mathbf{F}\mathbf{s}_{t-1} + \boldsymbol{\eta}_t, \quad \mathbf{F} = \begin{bmatrix} 1 & 0 \\ 0 & \rho_N \end{bmatrix}, \quad |\rho_N| < 1. \quad (29)$$

$$\boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}), \quad \mathbf{Q} = \begin{bmatrix} \sigma_{\tilde{N}}^2 & 0 \\ 0 & \sigma_{\hat{N}}^2 \end{bmatrix}. \quad (30)$$

The joint observation vector and measurement disturbances are

$$\mathbf{y}_t = \begin{bmatrix} \pi_t \\ n_t^{\text{CIQ}} \end{bmatrix}, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R}), \quad \mathbf{R} = \begin{bmatrix} \sigma_{\pi}^2 & 0 \\ 0 & \sigma_{\nu, N}^2 \end{bmatrix}. \quad (31)$$

For compact notation, define

$$\mathbf{d}_t = \begin{bmatrix} \alpha E_t \pi_{t+1} + \kappa_0 x_t \\ 0 \end{bmatrix}. \quad (32)$$

2.1 Model 0: CES

The CES benchmark imposes a constant Phillips-curve slope and excludes any direct competition channel.

$$\mathbf{y}_t = \mathbf{d}_t + \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \mathbf{s}_t + \boldsymbol{\varepsilon}_t, \quad \delta = \theta_0 = \gamma = 0. \quad (33)$$

2.2 Model 1: HSA Competition Effect on Slope

Model 1 allows the slow-moving competitive environment to change the Phillips-curve slope.

$$\kappa_t = \kappa_0 + \delta \tilde{N}_t, \quad \mathbf{y}_t = \mathbf{d}_t + \begin{bmatrix} \delta x_t & 0 \\ 1 & 1 \end{bmatrix} \mathbf{s}_t + \boldsymbol{\varepsilon}_t. \quad (34)$$

The specification remains conditionally linear and Gaussian because x_t is observed.

2.3 Model 2: HSA Direct Competition Channel

Model 2 keeps the Phillips-curve slope constant but allows the stationary competition state to affect inflation directly.

$$\mathbf{y}_t = \mathbf{d}_t + \begin{bmatrix} 0 & \theta_0 \\ 1 & 1 \end{bmatrix} \mathbf{s}_t + \boldsymbol{\varepsilon}_t, \quad \delta = \gamma = 0. \quad (35)$$

2.4 Model 3: HSA Competition Dynamic Effect

Model 3 allows the loading on the stationary competition state to depend on the slow-moving competitive environment.

$$\theta_t = \theta_0 + \gamma \tilde{N}_t, \quad \delta = 0. \quad (36)$$

$$\mathbf{y}_t = \mathbf{d}_t + \mathbf{h}_3(\mathbf{s}_t) + \boldsymbol{\varepsilon}_t, \quad \mathbf{h}_3(\mathbf{s}_t) = \begin{bmatrix} \theta_0 \hat{N}_t + \gamma \tilde{N}_t \hat{N}_t \\ \tilde{N}_t + \hat{N}_t \end{bmatrix}. \quad (37)$$

Because the inflation equation contains the product $\tilde{N}_t \hat{N}_t$, the measurement equation is bilinear in the latent states.

2.5 Model 4: HSA Joint Competition Effect on Slope and Direct Competition Channel

Model 4 combines the competition-dependent slope and the state-dependent direct competition channel.

$$\kappa_t = \kappa_0 + \delta \tilde{N}_t, \quad \theta_t = \theta_0 + \gamma \tilde{N}_t. \quad (38)$$

$$\mathbf{y}_t = \mathbf{d}_t + \mathbf{h}_4(\mathbf{s}_t) + \boldsymbol{\varepsilon}_t, \quad \mathbf{h}_4(\mathbf{s}_t) = \begin{bmatrix} \delta x_t \tilde{N}_t + \theta_0 \hat{N}_t + \gamma \tilde{N}_t \hat{N}_t \\ \tilde{N}_t + \hat{N}_t \end{bmatrix}. \quad (39)$$

The competition-related contribution to inflation can be decomposed as

$$C_t^{\text{slope}} = \delta \tilde{N}_t x_t, \quad C_t^{\text{direct}} = \theta_0 \hat{N}_t, \quad C_t^{\text{interaction}} = \gamma \tilde{N}_t \hat{N}_t. \quad (40)$$

$$C_t^{\text{dynamic}} = C_t^{\text{direct}} + C_t^{\text{interaction}}, \quad C_t^{\text{HSA}} = C_t^{\text{slope}} + C_t^{\text{dynamic}}. \quad (41)$$

3 Case 2: Quarterly Joint Estimation with Interpolated Gustavo Competition Data

The annual Gustavo competition series is first interpolated to the quarterly frequency. The resulting quarterly series is treated as the observed competition measure in the joint state-space system.

$$\mathbf{s}_t = \begin{bmatrix} \tilde{N}_t \\ \hat{N}_t \end{bmatrix}, \quad \tilde{N}_t = \bar{N}_t - N_0, \quad n_t^{\text{G,PCHIP}} = N_t^{\text{G,PCHIP}} - N_0. \quad (42)$$

$$\mathbf{s}_t = \mathbf{F}\mathbf{s}_{t-1} + \boldsymbol{\eta}_t, \quad \mathbf{F} = \begin{bmatrix} 1 & 0 \\ 0 & \rho_N \end{bmatrix}, \quad |\rho_N| < 1. \quad (43)$$

$$\boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}), \quad \mathbf{Q} = \begin{bmatrix} \sigma_{\tilde{N}}^2 & 0 \\ 0 & \sigma_{\hat{N}}^2 \end{bmatrix}. \quad (44)$$

$$\mathbf{y}_t = \begin{bmatrix} \pi_t \\ n_t^{\text{G,PCHIP}} \end{bmatrix}, \quad \mathbf{d}_t = \begin{bmatrix} \alpha E_t \pi_{t+1} + \kappa_0 x_t \\ 0 \end{bmatrix}. \quad (45)$$

$$\boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R}), \quad \mathbf{R} = \begin{bmatrix} \sigma_\pi^2 & 0 \\ 0 & \sigma_{\nu,N}^2 \end{bmatrix}. \quad (46)$$

3.1 Model 0: CES

The CES benchmark excludes both HSA competition channels.

$$\mathbf{y}_t = \mathbf{d}_t + \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \mathbf{s}_t + \boldsymbol{\varepsilon}_t, \quad \delta = \theta_0 = \gamma = 0. \quad (47)$$

3.2 Model 1: HSA Competition Effect on Slope

The slow competition state shifts the conditional Phillips-curve slope.

$$\kappa_t = \kappa_0 + \delta \tilde{N}_t, \quad \mathbf{y}_t = \mathbf{d}_t + \begin{bmatrix} \delta x_t & 0 \\ 1 & 1 \end{bmatrix} \mathbf{s}_t + \boldsymbol{\varepsilon}_t. \quad (48)$$

3.3 Model 2: HSA Direct Competition Channel

The stationary competition state enters inflation directly.

$$\mathbf{y}_t = \mathbf{d}_t + \begin{bmatrix} 0 & \theta_0 \\ 1 & 1 \end{bmatrix} \mathbf{s}_t + \boldsymbol{\varepsilon}_t, \quad \delta = \gamma = 0. \quad (49)$$

3.4 Model 3: HSA Competition Dynamic Effect

The direct competition loading varies with the slow competitive environment.

$$\theta_t = \theta_0 + \gamma \tilde{N}_t, \quad \delta = 0. \quad (50)$$

$$\mathbf{y}_t = \mathbf{d}_t + \begin{bmatrix} \theta_0 \hat{N}_t + \gamma \tilde{N}_t \hat{N}_t \\ \tilde{N}_t + \hat{N}_t \end{bmatrix} + \boldsymbol{\varepsilon}_t. \quad (51)$$

3.5 Model 4: HSA Joint Competition Effect on Slope and Direct Competition Channel

The full HSA specification allows both competition channels to operate simultaneously.

$$\kappa_t = \kappa_0 + \delta \tilde{N}_t, \quad \theta_t = \theta_0 + \gamma \tilde{N}_t. \quad (52)$$

$$\mathbf{y}_t = \mathbf{d}_t + \begin{bmatrix} \delta x_t \tilde{N}_t + \theta_0 \hat{N}_t + \gamma \tilde{N}_t \hat{N}_t \\ \tilde{N}_t + \hat{N}_t \end{bmatrix} + \boldsymbol{\varepsilon}_t. \quad (53)$$

$$C_t^{\text{slope}} = \delta \tilde{N}_t x_t, \quad C_t^{\text{direct}} = \theta_0 \hat{N}_t, \quad C_t^{\text{interaction}} = \gamma \tilde{N}_t \hat{N}_t. \quad (54)$$

4 Case 3: Mixed-Frequency Joint Estimation with Gustavo Competition Data

The annual Gustavo competition measure is not interpolated. Instead, the observed annual value is assigned to Q4, while Q1–Q3 are left missing. The underlying competition process is nevertheless defined at the quarterly frequency and estimated jointly within the state-space model.

$$\mathbf{s}_t = \begin{bmatrix} \tilde{N}_t \\ \hat{N}_t \end{bmatrix}, \quad \tilde{N}_t = \bar{N}_t - N_0. \quad (55)$$

The quarterly latent competition states evolve according to

$$\mathbf{s}_t = \mathbf{F}\mathbf{s}_{t-1} + \boldsymbol{\eta}_t, \quad \mathbf{F} = \begin{bmatrix} 1 & 0 \\ 0 & \rho_N \end{bmatrix}, \quad |\rho_N| < 1. \quad (56)$$

$$\boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}), \quad \mathbf{Q} = \begin{bmatrix} \sigma_{\tilde{N}}^2 & 0 \\ 0 & \sigma_{\hat{N}}^2 \end{bmatrix}. \quad (57)$$

The observed annual competition series is

$$n_t^{\text{G,obs}} = \begin{cases} N_t^{\text{G}} - N_0, & t \in \mathcal{Q}_4, \\ \text{missing}, & t \in \mathcal{Q}_1 \cup \mathcal{Q}_2 \cup \mathcal{Q}_3. \end{cases} \quad (58)$$

Its measurement equation is

$$n_t^{\text{G,obs}} = \tilde{N}_t + \hat{N}_t + \nu_t^N, \quad \nu_t^N \sim \mathcal{N}(0, \sigma_{\nu, N}^2), \quad t \in \mathcal{Q}_4. \quad (59)$$

For Q1–Q3, the competition observation is missing and therefore contributes no measurement density. The quarterly latent states continue to evolve through the transition equation and are inferred jointly from the complete state-space system.

The inflation equation is observed every quarter. Define

$$d_t = \alpha E_t \pi_{t+1} + \kappa_0 x_t, \quad \varepsilon_t^\pi \sim \mathcal{N}(0, \sigma_\pi^2). \quad (60)$$

4.1 Model 0: CES

The CES benchmark excludes competition from the inflation equation.

$$\pi_t = d_t + \varepsilon_t^\pi, \quad \delta = \theta_0 = \gamma = 0. \quad (61)$$

Together with the competition measurement equation,

$$\begin{bmatrix} \pi_t \\ n_t^{\text{G,obs}} \end{bmatrix} = \begin{bmatrix} d_t \\ \tilde{N}_t + \hat{N}_t \end{bmatrix} + \begin{bmatrix} \varepsilon_t^\pi \\ \nu_t^N \end{bmatrix}, \quad t \in \mathcal{Q}_4. \quad (62)$$

For Q1–Q3 only the first measurement equation is observed.

4.2 Model 1: HSA Competition Effect on Slope

The slow-moving competition state changes the Phillips-curve slope.

$$\kappa_t = \kappa_0 + \delta \tilde{N}_t, \quad \pi_t = d_t + \delta x_t \tilde{N}_t + \varepsilon_t^\pi. \quad (63)$$

Thus, in Q4,

$$\begin{bmatrix} \pi_t \\ n_t^{\text{G,obs}} \end{bmatrix} = \begin{bmatrix} d_t + \delta x_t \tilde{N}_t \\ \tilde{N}_t + \hat{N}_t \end{bmatrix} + \begin{bmatrix} \varepsilon_t^\pi \\ \nu_t^N \end{bmatrix}. \quad (64)$$

In Q1–Q3, $n_t^{\text{G,obs}}$ remains missing while the latent quarterly competition states are propagated by the transition equation.

4.3 Model 2: HSA Direct Competition Channel

The stationary competition state enters inflation directly.

$$\pi_t = d_t + \theta_0 \hat{N}_t + \varepsilon_t^\pi, \quad \delta = \gamma = 0. \quad (65)$$

In Q4,

$$\begin{bmatrix} \pi_t \\ n_t^{\text{G,obs}} \end{bmatrix} = \begin{bmatrix} d_t + \theta_0 \hat{N}_t \\ \tilde{N}_t + \hat{N}_t \end{bmatrix} + \begin{bmatrix} \varepsilon_t^\pi \\ \nu_t^N \end{bmatrix}. \quad (66)$$

In Q1–Q3, quarterly inflation helps update the latent competition state through its direct loading even though the annual competition observation is missing.

4.4 Model 3: HSA Competition Dynamic Effect

The direct loading itself varies with the slow-moving competition state.

$$\theta_t = \theta_0 + \gamma \tilde{N}_t, \quad \pi_t = d_t + \theta_0 \hat{N}_t + \gamma \tilde{N}_t \hat{N}_t + \varepsilon_t^\pi. \quad (67)$$

In Q4,

$$\begin{bmatrix} \pi_t \\ n_t^{\text{G,obs}} \end{bmatrix} = \begin{bmatrix} d_t + \theta_0 \hat{N}_t + \gamma \tilde{N}_t \hat{N}_t \\ \tilde{N}_t + \hat{N}_t \end{bmatrix} + \begin{bmatrix} \varepsilon_t^\pi \\ \nu_t^N \end{bmatrix}. \quad (68)$$

Because $\tilde{N}_t \hat{N}_t$ enters the inflation equation, this specification has a bilinear measurement equation.

4.5 Model 4: HSA Joint Competition Effect on Slope and Direct Competition Channel

The full specification combines the slope and dynamic competition channels.

$$\kappa_t = \kappa_0 + \delta \tilde{N}_t, \quad \theta_t = \theta_0 + \gamma \tilde{N}_t. \quad (69)$$

$$\pi_t = d_t + \delta x_t \tilde{N}_t + \theta_0 \hat{N}_t + \gamma \tilde{N}_t \hat{N}_t + \varepsilon_t^\pi. \quad (70)$$

In Q4, the complete measurement system is

$$\begin{bmatrix} \pi_t \\ n_t^{\text{G,obs}} \end{bmatrix} = \begin{bmatrix} d_t + \delta x_t \tilde{N}_t + \theta_0 \hat{N}_t + \gamma \tilde{N}_t \hat{N}_t \\ \tilde{N}_t + \hat{N}_t \end{bmatrix} + \begin{bmatrix} \varepsilon_t^\pi \\ \nu_t^N \end{bmatrix}. \quad (71)$$

In Q1–Q3,

$$n_t^{\text{G,obs}} = \text{missing}, \quad (72)$$

so only the quarterly inflation measurement is used. The latent competition states are not interpolated beforehand; their quarterly paths are inferred inside the state-space model from the transition dynamics, the Q4 competition observations, and the quarterly inflation equation.

$$C_t^{\text{slope}} = \delta \tilde{N}_t x_t, \quad C_t^{\text{direct}} = \theta_0 \hat{N}_t, \quad C_t^{\text{interaction}} = \gamma \tilde{N}_t \hat{N}_t. \quad (73)$$

5 Case 4: Mixed-Frequency Joint Estimation with Gustavo Competition Data and Number of Establishments

The annual Gustavo competition measure is observed only in Q4 and is not interpolated. Quarterly movements between the annual observations are inferred within the state-space model using both the latent competition dynamics and quarterly changes in the number of establishments.

Let E_t denote the quarterly number of establishments and define the demeaned quarterly establishment-growth signal as

$$g_t^E = \Delta \log E_t - \overline{\Delta \log E}. \quad (74)$$

The establishment series is used as an auxiliary signal for short-run business-demography movements rather than as a direct measure of competition.

The latent competition state is

$$\mathbf{s}_t = \begin{bmatrix} \tilde{N}_t \\ \hat{N}_t \end{bmatrix}, \quad \tilde{N}_t = \bar{N}_t - N_0. \quad (75)$$

The quarterly transition equation is

$$\mathbf{s}_t = \mathbf{F}\mathbf{s}_{t-1} + \mathbf{B}g_t^E + \boldsymbol{\eta}_t, \quad \mathbf{F} = \begin{bmatrix} 1 & 0 \\ 0 & \rho_N \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ \lambda_E \end{bmatrix}. \quad (76)$$

Equivalently,

$$\tilde{N}_t = \tilde{N}_{t-1} + \eta_t^{\tilde{N}}, \quad \hat{N}_t = \rho_N \hat{N}_{t-1} + \lambda_E g_t^E + \eta_t^{\hat{N}}, \quad |\rho_N| < 1. \quad (77)$$

The state innovations satisfy

$$\boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}), \quad \mathbf{Q} = \begin{bmatrix} \sigma_{\tilde{N}}^2 & 0 \\ 0 & \sigma_{\hat{N}}^2 \end{bmatrix}. \quad (78)$$

Thus, ρ_N captures persistence in short-run competition dynamics, while λ_E determines how quarterly establishment fluctuations inform the latent competition state.

The annual Gustavo observation is retained only in Q4:

$$n_t^{\text{G,obs}} = \begin{cases} N_t^{\text{G}} - N_0, & t \in \mathcal{Q}_4, \\ \text{missing}, & t \in \mathcal{Q}_1 \cup \mathcal{Q}_2 \cup \mathcal{Q}_3. \end{cases} \quad (79)$$

At every observed Q4, the latent states are required to reproduce the Gustavo competition measure exactly:

$$n_t^{\text{G,obs}} = [1 \quad 1] \mathbf{s}_t = \tilde{N}_t + \hat{N}_t, \quad t \in \mathcal{Q}_4. \quad (80)$$

Hence, the annual observations act as exact anchors for the quarterly latent competition path. In Q1–Q3, no artificial competition observation is created; the missing quarterly values are inferred from the transition dynamics, establishment fluctuations, and the inflation equation.

For compact notation, define

$$d_t = \alpha E_t \pi_{t+1} + \kappa_0 x_t, \quad \varepsilon_t^\pi \sim \mathcal{N}(0, \sigma_\pi^2). \quad (81)$$

5.1 Model 0: CES

The CES benchmark uses establishment growth only to recover the latent competition path; competition does not enter the inflation equation.

$$\pi_t = d_t + \varepsilon_t^\pi, \quad \delta = \theta_0 = \gamma = 0. \quad (82)$$

In Q4, the joint measurement system is

$$\begin{bmatrix} \pi_t \\ n_t^{\text{G,obs}} \end{bmatrix} = \begin{bmatrix} d_t & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ \mathbf{s}_t \end{bmatrix} + \begin{bmatrix} \varepsilon_t^\pi \\ 0 \end{bmatrix}, \quad t \in \mathcal{Q}_4. \quad (83)$$

In Q1–Q3, the second observation is missing.

5.2 Model 1: HSA Competition Effect on Slope

The slow-moving competition state determines the time-varying Phillips-curve slope.

$$\kappa_t = \kappa_0 + \delta \tilde{N}_t, \quad \pi_t = d_t + \delta x_t \tilde{N}_t + \varepsilon_t^\pi. \quad (84)$$

In Q4,

$$\begin{bmatrix} \pi_t \\ n_t^{\text{G,obs}} \end{bmatrix} = \begin{bmatrix} d_t + \delta x_t \tilde{N}_t \\ \tilde{N}_t + \hat{N}_t \end{bmatrix} + \begin{bmatrix} \varepsilon_t^\pi \\ 0 \end{bmatrix}. \quad (85)$$

5.3 Model 2: HSA Direct Competition Channel

The stationary competition state directly affects quarterly inflation.

$$\pi_t = d_t + \theta_0 \hat{N}_t + \varepsilon_t^\pi, \quad \delta = \gamma = 0. \quad (86)$$

In Q4,

$$\begin{bmatrix} \pi_t \\ n_t^{\text{G,obs}} \end{bmatrix} = \begin{bmatrix} d_t + \theta_0 \hat{N}_t \\ \tilde{N}_t + \hat{N}_t \end{bmatrix} + \begin{bmatrix} \varepsilon_t^\pi \\ 0 \end{bmatrix}. \quad (87)$$

In Q1–Q3, both establishment growth and quarterly inflation provide information about the unobserved competition state.

5.4 Model 3: HSA Competition Dynamic Effect

The direct competition loading varies with the slow-moving competition environment.

$$\theta_t = \theta_0 + \gamma \tilde{N}_t, \quad \delta = 0. \quad (88)$$

$$\pi_t = d_t + \theta_0 \hat{N}_t + \gamma \tilde{N}_t \hat{N}_t + \varepsilon_t^\pi. \quad (89)$$

In Q4,

$$\begin{bmatrix} \pi_t \\ n_t^{\text{G,obs}} \end{bmatrix} = \begin{bmatrix} d_t + \theta_0 \hat{N}_t + \gamma \tilde{N}_t \hat{N}_t \\ \tilde{N}_t + \hat{N}_t \end{bmatrix} + \begin{bmatrix} \varepsilon_t^\pi \\ 0 \end{bmatrix}. \quad (90)$$

The interaction $\tilde{N}_t \hat{N}_t$ makes the inflation measurement equation bilinear in the latent states.

5.5 Model 4: HSA Joint Competition Effect on Slope and Direct Competition Channel

The full specification combines the competition-dependent slope and dynamic competition channels.

$$\kappa_t = \kappa_0 + \delta \tilde{N}_t, \quad \theta_t = \theta_0 + \gamma \tilde{N}_t. \quad (91)$$

$$\pi_t = d_t + \delta x_t \tilde{N}_t + \theta_0 \hat{N}_t + \gamma \tilde{N}_t \hat{N}_t + \varepsilon_t^\pi. \quad (92)$$

In Q4, the complete measurement system is

$$\begin{bmatrix} \pi_t \\ n_t^{\text{G,obs}} \end{bmatrix} = \begin{bmatrix} d_t + \delta x_t \tilde{N}_t + \theta_0 \hat{N}_t + \gamma \tilde{N}_t \hat{N}_t \\ \tilde{N}_t + \hat{N}_t \end{bmatrix} + \begin{bmatrix} \varepsilon_t^\pi \\ 0 \end{bmatrix}. \quad (93)$$

In Q1–Q3,

$$n_t^{\text{G,obs}} = \text{missing}, \quad \hat{N}_t = \rho_N \hat{N}_{t-1} + \lambda_E g_t^E + \eta_t^{\hat{N}}. \quad (94)$$

The latent quarterly competition path therefore combines three sources of information: persistence in competition, quarterly business-demography movements, and the inflation equation, while every observed Q4 value remains exactly anchored to the original annual Gustavo series.

$$C_t^{\text{slope}} = \delta \tilde{N}_t x_t, \quad C_t^{\text{direct}} = \theta_0 \hat{N}_t, \quad C_t^{\text{interaction}} = \gamma \tilde{N}_t \hat{N}_t. \quad (95)$$

6 How to Estimate the Model

The estimation strategy is Bayesian. The computational method is organized according to the structure of each model:

$$\text{CES : Gibbs Sampling,} \quad \text{HSA Models 1-2 : Gibbs Sampling + FFBS,} \quad (96)$$

$$\text{HSA Models 3-4 : Gibbs Sampling + Blocked Conditional FFBS.} \quad (97)$$

This strategy applies to all four data cases. Case 4 differs only in its state transition equation, which additionally uses quarterly establishment growth as an observed transition input.

6.1 CES Benchmark: Gibbs Sampling

The CES benchmark is

$$\pi_t = \alpha E_t \pi_{t+1} + \kappa_0 x_t + \varepsilon_t^\pi, \quad \varepsilon_t^\pi \sim \mathcal{N}(0, \sigma_\pi^2). \quad (98)$$

Define

$$\beta_0 = \begin{bmatrix} \alpha \\ \kappa_0 \end{bmatrix}, \quad \mathbf{z}_t = \begin{bmatrix} E_t \pi_{t+1} \\ x_t \end{bmatrix}. \quad (99)$$

Stacking over t gives

$$\boldsymbol{\pi} = \mathbf{Z}\beta_0 + \boldsymbol{\varepsilon}^\pi, \quad \boldsymbol{\varepsilon}^\pi \sim \mathcal{N}(\mathbf{0}, \sigma_\pi^2 \mathbf{I}). \quad (100)$$

With

$$\beta_0 \sim \mathcal{N}(\mathbf{b}_0, \mathbf{V}_0), \quad \sigma_\pi^2 \sim \text{IG}(a_{\pi,0}, b_{\pi,0}), \quad (101)$$

the posterior coefficient update is

$$\mathbf{V}_n^{-1} = \mathbf{V}_0^{-1} + \frac{1}{\sigma_\pi^2} \mathbf{Z}'\mathbf{Z}, \quad (102)$$

$$\mathbf{b}_n = \mathbf{V}_n \left(\mathbf{V}_0^{-1} \mathbf{b}_0 + \frac{1}{\sigma_\pi^2} \mathbf{Z}'\boldsymbol{\pi} \right), \quad (103)$$

$$\beta_0 \mid \cdot \sim \mathcal{N}(\mathbf{b}_n, \mathbf{V}_n). \quad (104)$$

The variance update is

$$\sigma_\pi^2 \mid \cdot \sim \text{IG} \left(a_{\pi,0} + \frac{T}{2}, b_{\pi,0} + \frac{1}{2}(\boldsymbol{\pi} - \mathbf{Z}\boldsymbol{\beta}_0)'(\boldsymbol{\pi} - \mathbf{Z}\boldsymbol{\beta}_0) \right). \quad (105)$$

6.2 HSA Models 1–2: Gibbs Sampling with FFBS

For Cases 1–3, the latent competition state is

$$\mathbf{s}_t = \begin{bmatrix} \tilde{N}_t \\ \hat{N}_t \end{bmatrix}, \quad (106)$$

with transition equation

$$\mathbf{s}_t = \mathbf{F}\mathbf{s}_{t-1} + \boldsymbol{\eta}_t, \quad \mathbf{F} = \begin{bmatrix} 1 & 0 \\ 0 & \rho_N \end{bmatrix}. \quad (107)$$

$$\boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}), \quad \mathbf{Q} = \begin{bmatrix} \sigma_N^2 & 0 \\ 0 & \sigma_{\hat{N}}^2 \end{bmatrix}. \quad (108)$$

For Case 4, establishment growth enters the state transition as an observed input:

$$\mathbf{s}_t = \mathbf{F}\mathbf{s}_{t-1} + \mathbf{B}g_t^E + \boldsymbol{\eta}_t, \quad \mathbf{B} = \begin{bmatrix} 0 \\ \lambda_E \end{bmatrix}. \quad (109)$$

Thus,

$$\tilde{N}_t = \tilde{N}_{t-1} + \eta_t^{\tilde{N}}, \quad \hat{N}_t = \rho_N \hat{N}_{t-1} + \lambda_E g_t^E + \eta_t^{\hat{N}}. \quad (110)$$

Models 1 and 2 are conditionally linear Gaussian.

For Model 1,

$$\pi_t = \alpha E_t \pi_{t+1} + \kappa_0 x_t + \delta x_t \tilde{N}_t + \varepsilon_t^\pi. \quad (111)$$

For Model 2,

$$\pi_t = \alpha E_t \pi_{t+1} + \kappa_0 x_t + \theta_0 \hat{N}_t + \varepsilon_t^\pi. \quad (112)$$

The complete state path is therefore drawn jointly by FFBS.

6.2.1 Forward Filtering

Write the conditional linear-Gaussian system as

$$\mathbf{s}_t = \mathbf{c}_t + \mathbf{F}_t \mathbf{s}_{t-1} + \mathbf{w}_t, \quad \mathbf{y}_t = \mathbf{d}_t + \mathbf{H}_t \mathbf{s}_t + \mathbf{v}_t, \quad (113)$$

with

$$\mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_t), \quad \mathbf{v}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_t). \quad (114)$$

If

$$\mathbf{s}_{t-1} \mid \mathbf{y}_{1:t-1} \sim \mathcal{N}(\mathbf{m}_{t-1}, \mathbf{C}_{t-1}), \quad (115)$$

then the prediction step is

$$\mathbf{a}_t = \mathbf{c}_t + \mathbf{F}_t \mathbf{m}_{t-1}, \quad \mathbf{P}_t = \mathbf{F}_t \mathbf{C}_{t-1} \mathbf{F}_t' + \mathbf{Q}_t. \quad (116)$$

The innovation is

$$\mathbf{v}_t^y = \mathbf{y}_t - \mathbf{d}_t - \mathbf{H}_t \mathbf{a}_t, \quad \mathbf{S}_t^y = \mathbf{H}_t \mathbf{P}_t \mathbf{H}_t' + \mathbf{R}_t. \quad (117)$$

The Kalman gain and filtering update are

$$\mathbf{K}_t = \mathbf{P}_t \mathbf{H}_t' (\mathbf{S}_t^y)^{-1}, \quad (118)$$

$$\mathbf{m}_t = \mathbf{a}_t + \mathbf{K}_t \mathbf{v}_t^y, \quad \mathbf{C}_t = \mathbf{P}_t - \mathbf{K}_t \mathbf{S}_t^y \mathbf{K}_t'. \quad (119)$$

For mixed-frequency Cases 3 and 4, the competition observation is missing in Q1–Q3. The missing measurement row is simply omitted from the Kalman update. No quarterly competition observation is interpolated.

6.2.2 Backward Simulation

At the final period,

$$\mathbf{s}_T \sim \mathcal{N}(\mathbf{m}_T, \mathbf{C}_T). \quad (120)$$

For $t = T - 1, \dots, 1$,

$$\mathbf{J}_t = \mathbf{C}_t \mathbf{F}_{t+1}' \mathbf{P}_{t+1}^{-1}. \quad (121)$$

Then

$$\mathbf{s}_t \mid \mathbf{s}_{t+1}, \mathbf{y}_{1:T} \sim \mathcal{N}(\mathbf{m}_t + \mathbf{J}_t(\mathbf{s}_{t+1} - \mathbf{a}_{t+1}), \mathbf{C}_t - \mathbf{J}_t \mathbf{P}_{t+1} \mathbf{J}_t'). \quad (122)$$

6.3 HSA Models 3–4: Gibbs Sampling with Blocked Conditional FFBS

Models 3 and 4 contain the bilinear term

$$\gamma \tilde{N}_t \hat{N}_t. \quad (123)$$

The complete two-dimensional system is therefore not jointly linear in the latent states. Conditional on either latent state path, however, the other state enters linearly.

For Model 3,

$$\pi_t = \alpha E_t \pi_{t+1} + \kappa_0 x_t + \theta_0 \hat{N}_t + \gamma \tilde{N}_t \hat{N}_t + \varepsilon_t^\pi. \quad (124)$$

Conditional on $\hat{N}_{1:T}$,

$$\pi_t - \alpha E_t \pi_{t+1} - \kappa_0 x_t - \theta_0 \hat{N}_t = \gamma \tilde{N}_t \hat{N}_t + \varepsilon_t^\pi. \quad (125)$$

Conditional on $\tilde{N}_{1:T}$,

$$\pi_t - \alpha E_t \pi_{t+1} - \kappa_0 x_t = (\theta_0 + \gamma \tilde{N}_t) \hat{N}_t + \varepsilon_t^\pi. \quad (126)$$

For Model 4,

$$\pi_t = \alpha E_t \pi_{t+1} + \kappa_0 x_t + \delta x_t \tilde{N}_t + \theta_0 \hat{N}_t + \gamma \tilde{N}_t \hat{N}_t + \varepsilon_t^\pi. \quad (127)$$

Conditional on $\hat{N}_{1:T}$,

$$\pi_t - \alpha E_t \pi_{t+1} - \kappa_0 x_t - \theta_0 \hat{N}_t = (\delta x_t + \gamma \hat{N}_t) \tilde{N}_t + \varepsilon_t^\pi. \quad (128)$$

Conditional on $\tilde{N}_{1:T}$,

$$\pi_t - \alpha E_t \pi_{t+1} - \kappa_0 x_t - \delta x_t \tilde{N}_t = (\theta_0 + \gamma \tilde{N}_t) \hat{N}_t + \varepsilon_t^\pi. \quad (129)$$

Hence the Gibbs sampler alternates between two exact conditional FFBS draws:

$$\tilde{N}_{1:T}^{(g)} \sim p\left(\tilde{N}_{1:T} \mid \hat{N}_{1:T}^{(g-1)}, \boldsymbol{\Theta}^{(g-1)}, \mathcal{D}\right), \quad (130)$$

$$\hat{N}_{1:T}^{(g)} \sim p\left(\hat{N}_{1:T} \mid \tilde{N}_{1:T}^{(g)}, \boldsymbol{\Theta}^{(g-1)}, \mathcal{D}\right). \quad (131)$$

The same blocked conditional FFBS strategy is used in Case 4. The establishment-growth term simply enters the conditional transition mean of $\hat{N}_t$.

6.4 Transition-Parameter Gibbs Updates

For Cases 1–3,

$$\hat{N}_t = \rho_N \hat{N}_{t-1} + \eta_t^{\hat{N}}. \quad (132)$$

For Case 4,

$$\hat{N}_t = \rho_N \hat{N}_{t-1} + \lambda_E g_t^E + \eta_t^{\hat{N}}. \quad (133)$$

Thus Case 4 can be written as the Gaussian regression

$$\mathbf{r}_E = \mathbf{X}_E \begin{bmatrix} \rho_N \\ \lambda_E \end{bmatrix} + \boldsymbol{\eta}^{\hat{N}}, \quad (134)$$

where

$$\mathbf{r}_E = \begin{bmatrix} \hat{N}_2 \\ \vdots \\ \hat{N}_T \end{bmatrix}, \quad \mathbf{X}_E = \begin{bmatrix} \hat{N}_1 & g_2^E \\ \vdots & \vdots \\ \hat{N}_{T-1} & g_T^E \end{bmatrix}. \quad (135)$$

With a Gaussian prior, (ρ_N, λ_E) has a Gaussian conditional posterior, truncated only to enforce $|\rho_N| < 1$. The state residuals are

$$u_t^{\hat{N}} = \hat{N}_t - \tilde{N}_{t-1}, \quad (136)$$

and

$$u_t^{\hat{N}} = \begin{cases} \hat{N}_t - \rho_N \hat{N}_{t-1}, & \text{Cases 1–3,} \\ \hat{N}_t - \rho_N \hat{N}_{t-1} - \lambda_E g_t^E, & \text{Case 4.} \end{cases} \quad (137)$$

The corresponding innovation variances are updated from inverse-gamma full conditionals.

6.5 Complete Gibbs Sweep

For HSA Models 1 and 2:

1. Draw the complete state path $\mathbf{s}_{1:T}$ by FFBS.
2. Draw the inflation coefficients conditional on the state path.
3. Draw σ_π^2 .
4. Draw the transition parameters ρ_N and, in Case 4, λ_E .
5. Draw the state-innovation variances and applicable measurement variance.
6. Store the post-warm-up draw.

For HSA Models 3 and 4:

1. Draw $\tilde{N}_{1:T}$ by FFBS conditional on $\hat{N}_{1:T}$.
2. Draw $\hat{N}_{1:T}$ by FFBS conditional on $\tilde{N}_{1:T}$.
3. Draw the inflation coefficients conditional on both state paths.
4. Draw σ_π^2 .
5. Draw the transition parameters ρ_N and, in Case 4, λ_E .
6. Draw the state-innovation variances and applicable measurement variance.
7. Store the post-warm-up draw.

6.6 Convergence and Computational Validation

All models are estimated using multiple dispersed chains. Convergence is assessed using

$$\hat{R}, \quad ESS_{\text{bulk}}, \quad ESS_{\text{tail}}, \quad (138)$$

together with trace plots and autocorrelation summaries.

For Models 1 and 2, the full two-dimensional FFBS is used directly. For Models 3 and 4, the blocked conditional FFBS implementation is validated against the linear special case obtained by setting $\gamma = 0$.

For mixed-frequency Cases 3 and 4, Q1–Q3 remain missing throughout the estimation. The quarterly latent competition path is inferred by the state-space smoother rather than constructed by deterministic interpolation.